Thinesh Thiyakesan Ponbagavathi

thinesh516@gmail.com github.com/th-nesh linkedin.com/in/thinesh-tp Stuttgart, Germany

Education

University of Stuttgart, Germany 2024 – Present

PhD in Computer Science advised by Prof. Dr. Alina Roitberg
Scholar at the International Max Planck Research School for Intelligent Systems (IMPRS-IS)

University of Stuttgart, Germany 2021 – 2023

M.Sc. (1.6/1.0) in Electrical Engineering (Smart Systems)
Thesis: 3D-Aware Urban Scene Generation advised by Prof. Dr. Bin Yang

Anna University, India 2014 – 2018

B.E. (8.63/10) in Electronics and Instrumentation Engineering
Thesis: Cognitive Smart Cities using LoRa advised by Prof. Dr. C. Esakkiappan

Research Interests

Computer Vision · Vision Foundation Models · Video Understanding · Parameter-Efficient Fine-Tuning (PEFT) · View-Invariant Representation Learning · Human-Robot Interaction

Experience

PhD Researcher, **University of Stuttgart (IntCDC) & University of Hildesheim** 2024 – Present

Develop parameter-efficient adaptations of vision foundation models for fine-grained video understanding.
Design cross-view and temporal modeling strategies for robust recognition of nearly symmetric actions.

Lead Controls Engineer, **Johnson Controls** 2018 – 2021

Designed control systems for HVAC and building automation across large-scale projects.

Selected Publications

- *Frame2Freq: Spectral Adapters for Fine-Grained Video Understanding.*
  Thinesh Thiyakesan Ponbagavathi, Constantin Seibold and Alina Roitberg
  IEEE/CVF Conf. on Computer Vision and Pattern Recognition (***CVPR***), 2026.

- *Not All Layers are Equal for Image-to-Video Transfer.*
  Thinesh Thiyakesan Ponbagavathi, Constantin Seibold and Alina Roitberg
  Paper Under Review, 2026.

- *Order Matters: On Parameter-Efficient Image-to-Video Probing for Recognizing Nearly Symmetric Actions.*
  Thinesh Thiyakesan Ponbagavathi and Alina Roitberg
  IEEE Int. Conf. on Robotics and Automation (***ICRA***), 2026.

- *T-MASK: Temporal Masking for Probing Foundation Models across Camera Views in Driver Monitoring.*
  Thinesh Thiyakesan Ponbagavathi, Kunyu Peng and Alina Roitberg
  IEEE Int. Conf. on Intelligent Transportation Systems (***ITSC***), 2025. ***Best Student Paper Finalist.***

Academic Activities

Selected Participant, International Computer Vision Summer School (ICVSS), Sicily	2026
Supervisor for (5×) M.Sc. Theses	2024 – Present
Teaching Assistant, Automotive and Assistive Computer Vision	2025
Oral Presentation, High Performance Computing in Science & Engineering, HLRS Workshop	2025
Lead Organizer, Seminar on Limitations of Foundation Models, Hildesheim	2026 – Present
Lead Organizer, Lab Course on Multimodal Recognition & Deep Learning, Hildesheim	2026 – Present
Reviewer: IEEE IV (2025), ITSC (2025), ICRA (2026)	2025 – 2026

Skills

Programming Languages:	Python, C++
Frameworks & Libraries:	PyTorch, TensorFlow, HuggingFace, OpenCV
Tools:	Git, SLURM, LaTeX, Distributed Training (DDP)
Languages:	Tamil (Native), English (Full Professional Proficiency), German (Elementary)

Awards and Honors

Best Student Paper Finalist, IEEE Int. Conf. on Intelligent Transportation Systems (ITSC)	2025
PhD Scholar, International Max Planck Research School for Intelligent Systems (IMPRS-IS)	2024
Rank 2/22, M.Sc. Electrical Engineering, University of Stuttgart	2023
University Rank Medal, Anna University (Rank 18/1079)	2018
Department Topper Cash Prize (3×) for topping semester examinations, Anna University	2014–2018

Additional Publications

- *Structured Relational Reasoning for Group Activity Assessment.*
  Thinesh Thiyakesan Ponbagavathi*, Chengzheng Yang* and Alina Roitberg (*Equal contribution)
  *IEEE/CVF Conf. on Computer Vision and Pattern Recognition Workshops (**CVPRW**)*, 2026.

- *PDNet: Exploring Pruning and Distillation for Real-Time Worker Activity Recognition in Industrial Assembly.*
  Malek Baba, **Thinesh Thiyakesan Ponbagavathi** and Alina Roitberg
  *Int. Conf. on Computer Vision Theory and Applications (**VISAPP**)*, 2026. ***Oral Presentation.***

- *SceneGraphRAG: Efficient and Scalable Object Grounding Using Knowledge-Enhanced 3D Scene Graphs.*
  Michael Pabst, **Thinesh Thiyakesan Ponbagavathi**, Deniz Bickici, Dennis Rotondi, Shohei Mori, Kai O. Arras and Dieter Schmalstieg
  Paper Under Review, 2026.

- *RealityGraph: Generating, Editing and Applying Semantic Scene Graphs in Augmented Reality.*
  Michael Pabst, **Thinesh Thiyakesan Ponbagavathi**, Jan Kolberg, Verena Biener, Shohei Mori, Alina Roitberg and Dieter Schmalstieg
  Paper Under Review, 2025.